

SOURDOUGH BREAD

YIELD: 2 loaves (~900 g each)

February 23, 2026 · Version 1 · Side Dish

Lean naturally leavened bread with staged fermentation, folds, and covered baking.

PREPARATION

Scaling basis: Bread flour · 1100 g

INGREDIENT	QUANTITY	SCALING	PROCEDURE
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PHASE A — BUILD LEVAIN

<u>Mature sourdough starter</u>	20 g	1.82%	1. Mix starter, flour, and water until homogeneous.
Inoculate the levain.			2. Cover and ferment until expanded and active.
<u>Bread flour</u>	100 g	9.09%	
Feed the levain.			
<u>Water</u>	100 g	9.09%	
Hydrate the levain.			

TECHNIQUE APPLICATIONS

Technique	Controls	Purpose
<u>Levain build</u>	Ferment about 8 h at room temperature until expanded and active.	Produce ripe leaven.

PRINCIPLES

- Yeast and lactic acid bacteria fermentation — The culture produces gas and acidity.

PHASE OUTPUTS

Key	Phase Output	Description
ripe-levain	Ripe levain	Active leaven for the dough.

PHASE B — MIX AND DEVELOP DOUGH

<u>Bread flour</u>	1000 g	90.91%	3. Combine water, ripe levain, and flour; cover and rest.
Mix into the dough.			

<u>Water</u>	640 g	58.18%	4. Dissolve salt in the additional water, mix it into the dough, and perform three stretch-and-fold sets.
Use 600 g to hydrate the flour and reserve 40 g to dissolve salt.			5. Let the dough bulk ferment until visibly aerated.
<u>Salt</u>	20 g	1.82%	
Dissolve in the additional water.			

TECHNIQUE APPLICATIONS

Technique	Controls	Purpose
<u>Autolyse</u>	Rest flour and water 30 min to 4 h.	Hydrate flour before salt incorporation.
<u>Folding</u>	Perform 3 stretch-and-fold sets over 90 min.	Strengthen dough without intensive kneading.
<u>Bulk fermentation</u>	Ferment 2–3 h until visibly aerated.	Develop gas and dough structure.

PRINCIPLES

- Hydration ratio — Water proportion controls dough handling and crumb.
- Gluten development — Resting and folding develop structure.
- Salt control of fermentation — Salt affects fermentation timing.

PHASE OUTPUTS USED

Phase Output	Use
Ripe levain	Dissolve ripe-levain in the dough water.

PHASE OUTPUTS

Key	Phase Output	Description
bulk-dough	Bulk-fermented dough	Aerated dough ready to divide.

PHASE C — SHAPE, PROOF, AND BAKE

6. Divide the dough into two, pre-shape, rest 30 min, shape, and proof 2–3 h at room temperature or overnight in the refrigerator.
7. Preheat a lidded casserole at 240 C for at least 20 min.

8. Bake each scored loaf covered, uncover for the final 10 min, and cool completely before slicing.

TECHNIQUE APPLICATIONS

Technique	Controls	Purpose
<u>Staged baking</u>	Bake covered then uncover for the last 10 min; total about 40 min at 240 C.	Allow spring before browning the crust.

PRINCIPLES

- Steam-assisted oven spring – Covered baking delays crust setting during expansion.
- Maillard reaction – The uncovered finish browns the crust.

PHASE OUTPUTS USED

Phase Output	Use
Bulk-fermented dough	Divide bulk-dough , pre-shape, rest, shape, proof, and score.

FAILURE MODES

Symptom	Likely cause	Corrective action
Dense loaf	Levain or bulk fermentation was insufficient.	Extend fermentation and confirm levain activity.
Flat dough	Gluten was weak or dough overproofed.	Add a fold set and shorten final proof.
Gummy crumb	Centre was underbaked or sliced hot.	Bake to about 98 C and cool fully.